

ARAVIND GOUD KALAALI

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OBJECTIVE

Associate Data Scientist with hands-on experience in Python, Data Analysis, and specialized expertise in statistical modeling, Deep Learning, NLP, and Machine Learning. Skilled in managing large datasets, deriving actionable insights through data visualization, and collaborating with cross-functional teams to deliver data-driven recommendations. Seeking a full-time role to apply expertise in developing innovative solutions that enhance decision-making and drive business growth. Currently exploring Generative AI to enhance solutions.

EDUCATION

Master in Data Science (GPA: 3.66/4)

University of Massachusetts, Dartmouth, USA

2023 - 2025

B.Tech in Electronics and Communication Engineering, (GPA: 3.2/4)

Central University of Karnataka, Karnataka, India

2018 - 2022

RELEVANT COURSES

Python for Data Science	Mathematical Statistics	Advanced Machine Learning	Deep Learning
Advanced Data Mining	Database Design	Data Visualization	Natural Language Processing
Flask	Computer Vision	SQL	Applied Business Analytics

EXPERIENCE

Associate Data Scientist

HBIC Solutions

Feb 2023 - July 2023

Hyderabad, India

- Developed predictive models employing Machine Learning, Natural Language processing and Deep Learning Algorithms to address business challenges. Refined data quality by implementing rigorous data cleaning, pre-processing, and handling missing values and outliers, boosting data reliability by 95% , and contributing to more accurate and actionable insights.
- Collaborated with cross-functional teams to understand business requirements into technical specifications and deliver data- driven solutions. Worked with a team of data scientists on multiple projects including Time Series Sales Forecasting, Image Classification, Sentiment Analysis etc.
- Authored more than 10+ comprehensive documentation for the data analysis process, models built, and results obtained.

Data Science Apprentice

Innomatics Research Labs

Jul 2022 - Feb 2023

Hyderabad, India

- Gained hands-on experience in data science techniques, including data preprocessing, statistical analysis, machine learning and deep learning model development .
- Managed data preprocessing and cleaning for large datasets, and visualized insights to guide team decisions.
- Strengthened skills in Python for Data Science, Machine Learning, Deep Learning, NLP, SQL, and data visualization tools like Tableau.

PROJECTS

Flight Fare Price Prediction. Scraped flight fare data, converting raw data into a structured format for analysis. Developed a machine learning model using AWS SageMaker and Python to predict flight fares. Implemented XGBoost and Random Forest, reducing Mean Absolute Error (MAE) by 18%. Improved real-time price estimation, enhancing revenue optimization and enabling seamless integration into dynamic pricing systems.

Exploratory Data Analysis. Scraped university data from Shiksha website and converted raw data into a structured format. Conducted Exploratory Data Analysis (EDA) to rank U.S. universities for MS programs, employing statistical methods to assist students in decision-making. Provided actionable insights through data visualization, facilitating informed choices for academic pursuits. Analyzed rankings using univariate and bivariate analysis, highlighting key factors influencing university selection for MS in Computer Science.

Big Mart Sales Prediction. Applied Linear Regression, Random Forest, and XGBoost to forecast sales, achieving an R^2 score of 0.89. Identified key sales trends, optimizing inventory management and revenue planning. The model reduced stock issues, ensuring better supply chain efficiency. Insights enabled data-driven product placement and marketing strategies.

CRUD Operations and Telephone Directory. Developed a Flask-based web application for CRUD operations using SQLite, streamlining contact data management. Enabled users to create, delete, and update entries efficiently in a telephone directory. Designed an intuitive interface for seamless user interaction and data accessibility. Improved data integrity and scalability for enhanced usability.

SMS Spam Detection Algorithm. Built an NLP-based spam classifier using TF-IDF, Logistic Regression and Naïve Bayes, achieving 98% accuracy. Enhanced communication security by filtering spam and protecting user privacy. The real-time model integration reduced fraudulent and intrusive messages. Continuous learning improved detection against evolving spam patterns.

CERTIFICATION

- Exploratory Data Analysis Using Python offered by Innomatics research labs, Python Programming with Application Development offered by Innomatics research labs, IBM Certified Data Science offered by Innomatics research labs. The Joy of Computing using python offered by IIT Madras.
- Industrial Training: Gained practical insights into IoT applications through online training with The National Small Industrial Corporation LTD.

SKILLS

Programming Language: Python, R Language.

Statistical Analysis: Hypothesis Testing, ANOVA, Chi-Square, Descriptive Analysis.

AI/ML frameworks and models: PyTorch, TensorFlow and Keras, Building and fine-tuning Large Language models, Computer Vision.

Machine Learning: Supervised and Unsupervised learning (Classification, Regression, Clustering, Dimensionality Technique), Ensemble Techniques.

Natural Language Processing: RNN, LSTM, Encoder-Decoder, BERT, Transformers

Deep Learning: Neural Networks (ANN, CNN).

Tools: MS Office (Excel, PowerPoint).

Databases / DBMS: MySQL.

Language Frameworks and Libraries: NumPy, Pandas, Scikit-Learn, ggplot2.

Data Visualization: Tableau, Matplotlib, Seaborn, Plotpy.

Operating Systems: MacOS, Windows.

EXTRA- CURRICULAR ACTIVITIES

- Secured 2nd place with a project on "Bluetooth-based Home Automation" at the Central University of Karnataka.
- Actively participated in the SYWLC, as well as various webinars and seminars organized by the IEEE Bangalore Section.